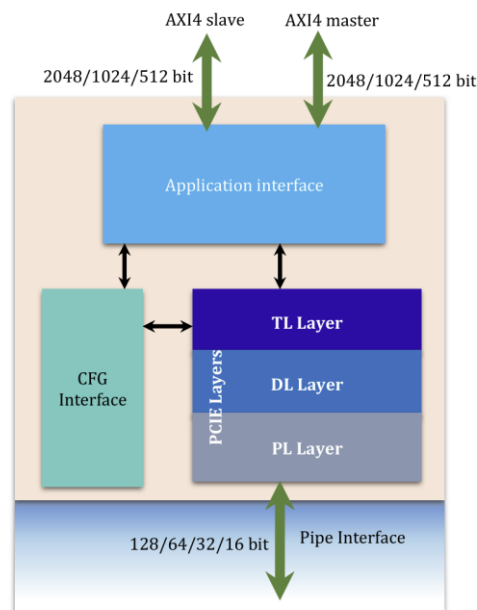


## PCIe Gen 5 Controller IP

With flexible multi-lane connection configurations and a signalling rate of 32.0 GT/s per lane, PCI Express Gen5 (PCIe 5.x) is a high-speed, point-to-point serial interconnect standard that enables incredibly high throughput for networking, storage, and accelerator applications. The PCIe specification's three primary layers—Transaction, Data Link, and Physical (digital packet and link control)—are implemented using this IP. The IP also includes PIPE interface logic to communicate with a PCIe Gen5 PHY and AXI bridge logic to interface with high-performance system fabrics.



### Features

- A maximum data throughput of 32 GT/s per lane is supported by the interface.
- Root Port, Endpoint, Switch Port, Dual-Mode operation, and Retimer-based configurations are among the topologies that are supported.
- For widths of x1, x2, x4, x8, and x16, lane layouts are supported.
- PCIe 4.0 and PCIe 3.1/3.0 standards are backward compatible with this configuration.
- Features related to Reliability, Availability, and Serviceability (RAS) are supported.
- There is support for both FLIT-based and non-FLIT virtual channel topologies.
- To enhance data integrity, Forward Error Correction (FEC) is supported.
- For flexible board routing, Lane Orientation Polarity (LOP) support is offered

### Application

- Consumer Electronics.
- Defence & Aerospace.
- Virtual Reality.
- Augmented Reality.
- Medical.
- Biometrics (Fingerprints, etc).
- Automotive Devices.
- Sensor Devices.

### Deliverables

- Verilog Source code.
- User Guide.
- IP Integration Guide.
- Run and Synthesis script.
- Encrypted Verification Testbench Environment.